MOS Capacitors

Streetman & Banerjee, Chapter 6:
MOS Capacitors

Streetman and Banerjee, 'Solid State Electronic Devices,' 6^{th} ed., Pearson, 2006.

1

Questions?

2

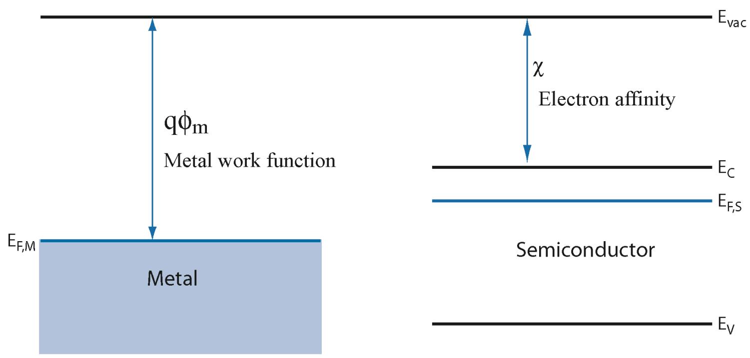
Review: Metal-Semiconductor Junctions

1. Schottky diode
    1. Metal workfunction
    2. Electron affinity
    3. Schottky barrier height
    4. Band alignments
    5. Electrostatics (Depletion width, capacitance)

3

Metal-Semiconductor Junction (Schottky Diode)

- Energies and alignments from the vacuum level:



- Metal workfunction Φ_m: distance from the vacuum level to the metal Fermi level.
- Electron affinity χ: distance from the semiconductor conduction band edge to the vacuum. This value is a property of the material (Si), and it is, therefore, the same at any point in the band diagram.
- The vacuum level, E_{vac}, is a continuous function of position (no discrete jumps, no step functions).

4

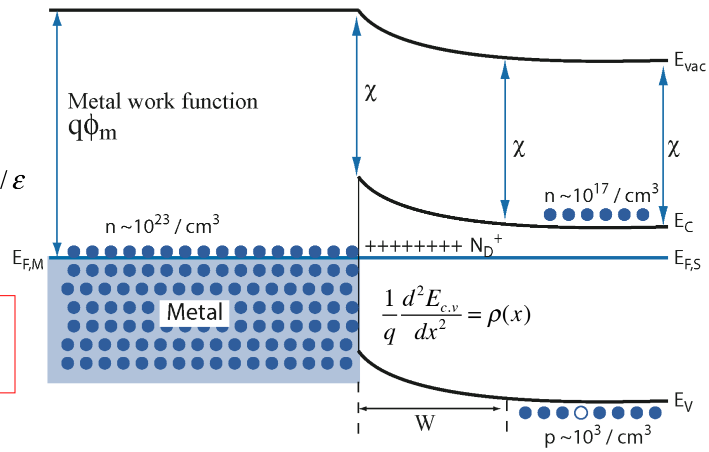
Metal-Semiconductor Junction Potential

- Band curvature

$$\nabla \cdot \mathcal{E} = \rho(\mathbf{r})/\varepsilon$$

$$\nabla \cdot (-\nabla V) = \rho(\mathbf{r})/\varepsilon$$

$$-\frac{d^2V}{dx^2} = \frac{1}{q}\boxed{\frac{d^2E_c}{dx^2} = \rho(x)}/\varepsilon$$



The sign of the second derivative is the sign of the charge.

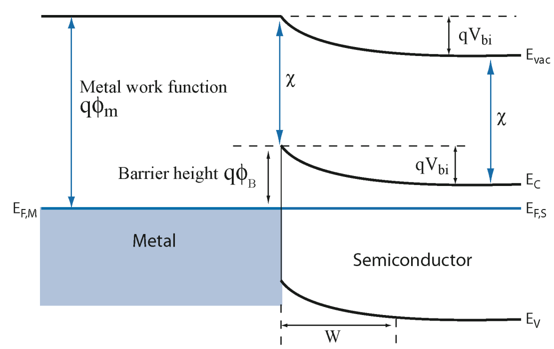
- In a region of positive charge, the conduction and valence band are 'concave up.' The second derivative of the conduction band or valence band with repect to x is positive.
- In a region of negative charge, the bands are 'concave down.' The second derivatives are negative.
- If there is no curvature (if the bands are linear), there is no charge.

5

Metal-Semiconductor Junction (Schottky Diode)

- Schottky Barrier height Φ_B:

$$q\phi_B = q\phi_m - \chi$$



- Built-in voltage V_{bi}:

$$qV_{bi} = q\phi_B - (E_c - E_F)_{\text{bulk}} = q\phi_m - \chi - (E_c - E_F)_{\text{bulk}}$$

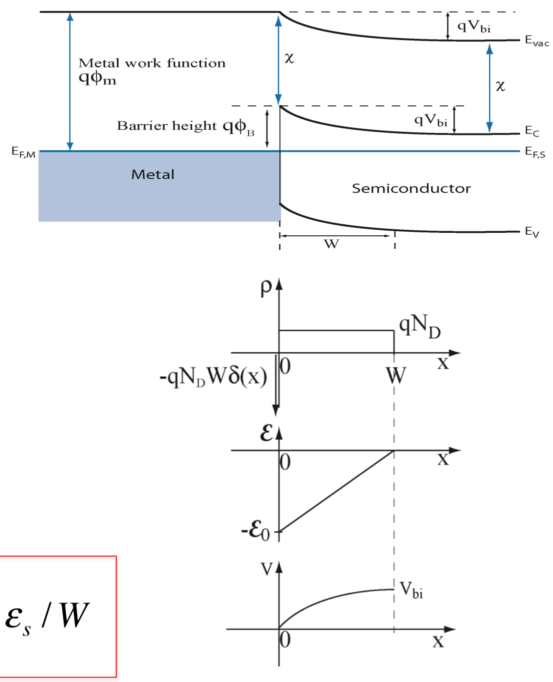
$$n = N_D = N_C e^{-(E_C - E_F)/kT} \Rightarrow -(E_C - E_F) = kT\ln\left(\frac{N_D}{N_C}\right)$$

$$qV_{bi} = q\phi_m - \chi + kT\ln\left(\frac{N_D}{N_C}\right)$$

6

Metal-Semiconductor Junction Electrostatics

- Depletion Width, W
- Capacitance, ε_s / W.
- Just like one-sided pn junction.



$$qV_{bi} = q\phi_m - \chi + kT\ln\left(\frac{N_D}{N_C}\right)$$

$$\mathcal{E}_0 = \frac{qN_D W}{\varepsilon_s}$$

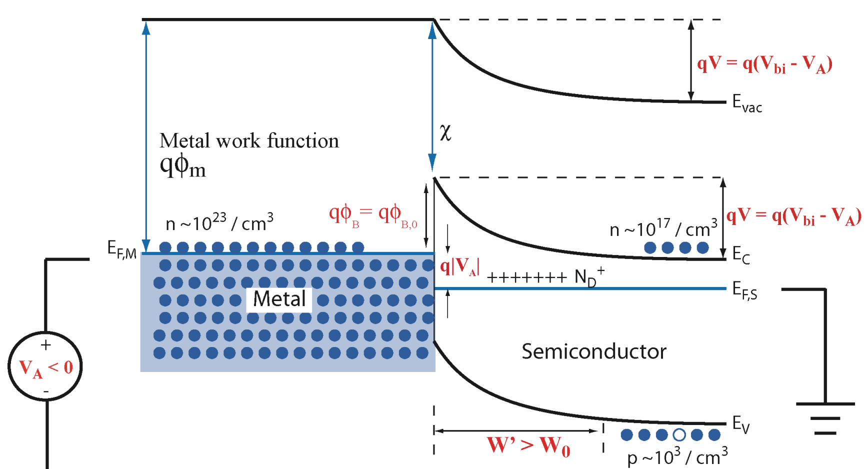
$$V_{bi} = \frac{1}{2}\mathcal{E}_0 W = \frac{qN_D W^2}{2\varepsilon_s}$$

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}}$$

$$C_j = \varepsilon_s / W$$

7

Schottky Diode Under REVERSE Bias



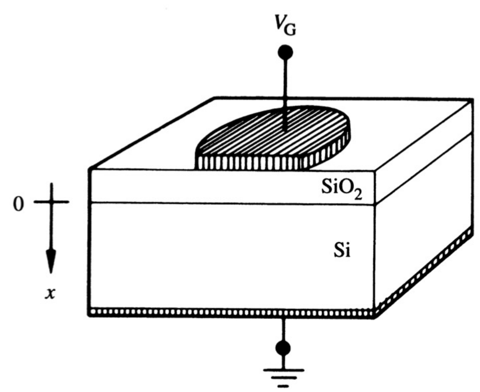
$$W = \sqrt{\frac{2\varepsilon_s (V_{bi} - V_A)}{qN_D}}$$

- The metal and semiconductor quasi-Fermi levels split at the metal-semiconductor junction by the magnitude of the applied bias, $q|V_A|$.
- The Schottky barrier height is unchanged.
- The voltage drop across the semiconductor increases to $V_{bi} - V_A$, just like in a pn junction.
- The depletion width W increases.

8

Metal-Oxide-Semiconductor (MOS) Capacitor
Ch. 6

1. Band lineups, band diagram
2. Surface potential

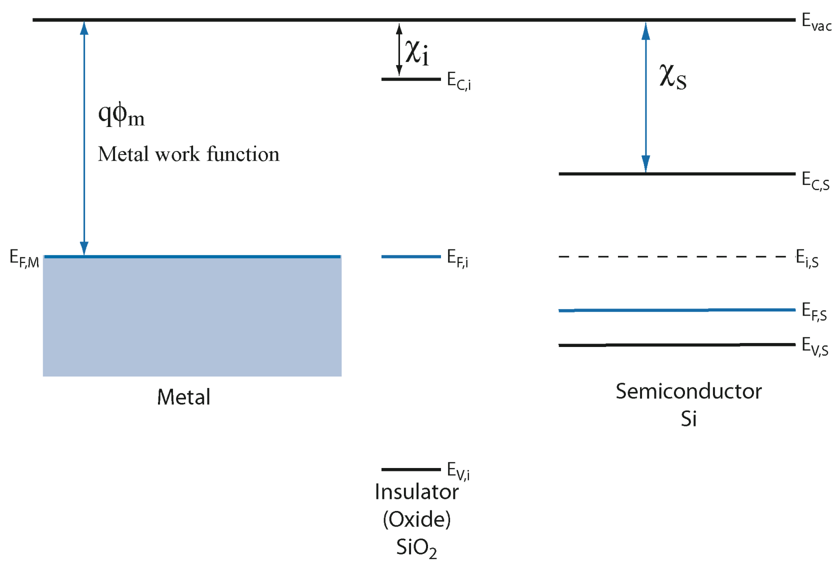


Looks like a Schottky diode but with SiO_2 between the metal and the semiconductor.

9

Metal-Oxide-Semiconductor (MOS) Capacitor

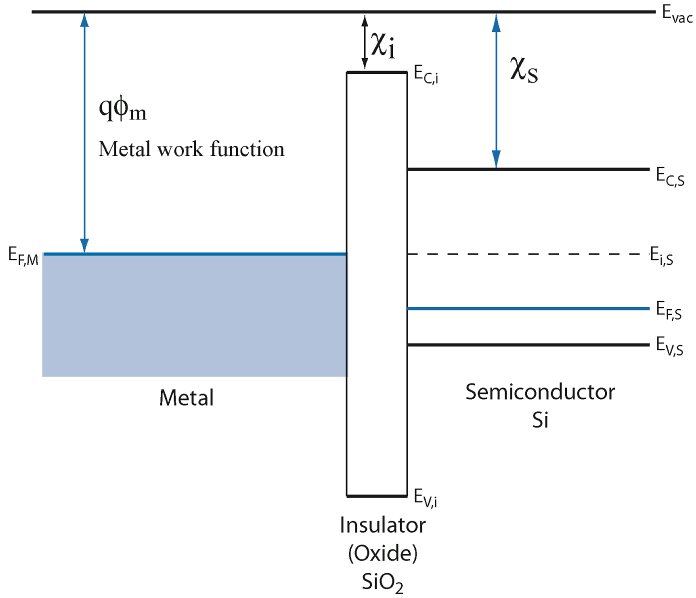
1. Band lineups



10

Metal-Oxide-Semiconductor (MOS) Capacitor

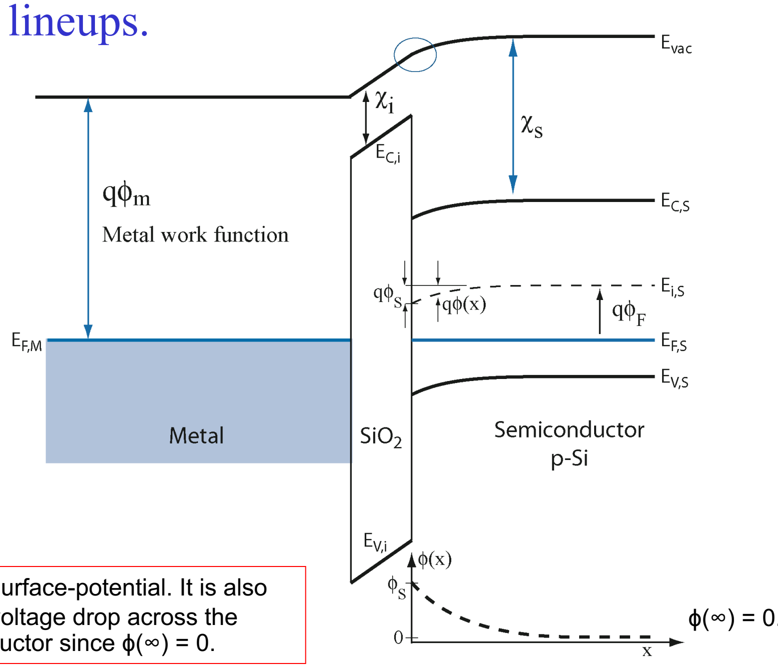
1. Band lineup at t=0.



11

Metal-Oxide-Semiconductor (MOS) Capacitor

1. Band lineups.



ϕ_s is the surface-potential. It is also the total voltage drop across the semiconductor since $\phi(\infty) = 0$.

12

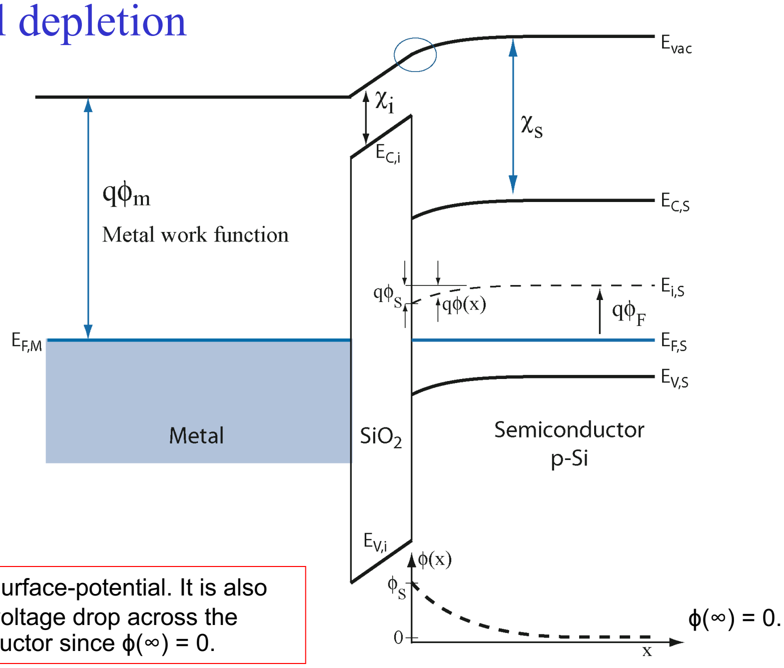
MOS Bias Conditions

1. Accumulation. When the density of majority carriers is increased at the surface.
2. Flat band. When the bands are flat in the semiconductor.
3. Depletion. When the density of majority carriers is depleted at the surface.
4. Inversion. When the carrier type is inverted at the interface.

13

MOS Equilibrium

1. Small depletion

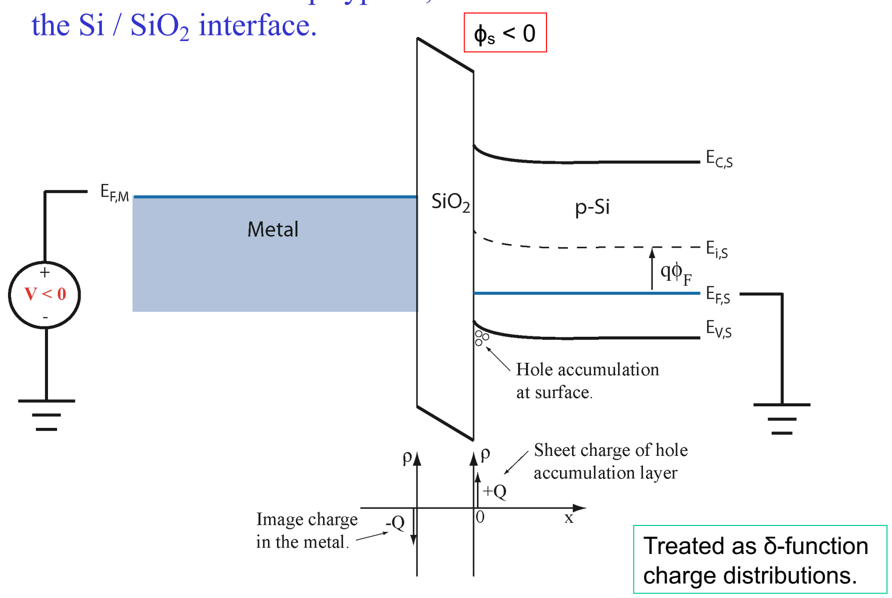


ϕ_s is the surface-potential. It is also the total voltage drop across the semiconductor since $\phi(\infty) = 0$.

14

MOS Accumulation

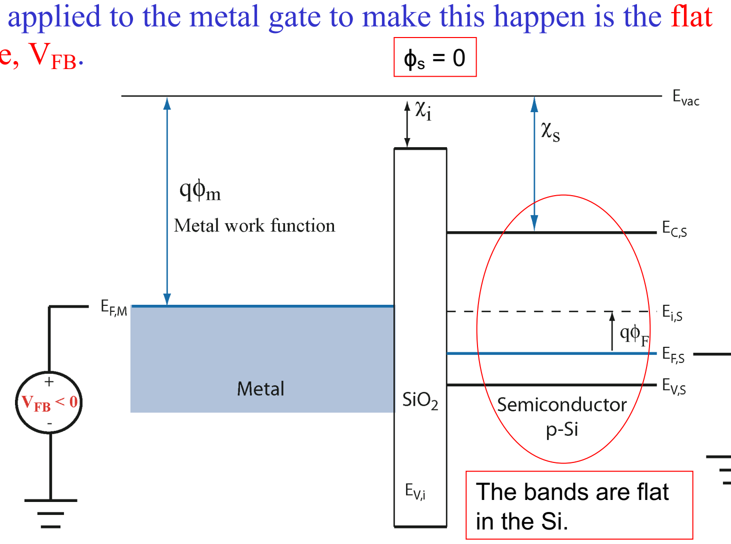
1. Accumulation is when the majority carrier density is enhanced at the surface of the Si. For p-type Si, this means that holes are accumulated at the Si / SiO_2 interface.



15

MOS Flat Band

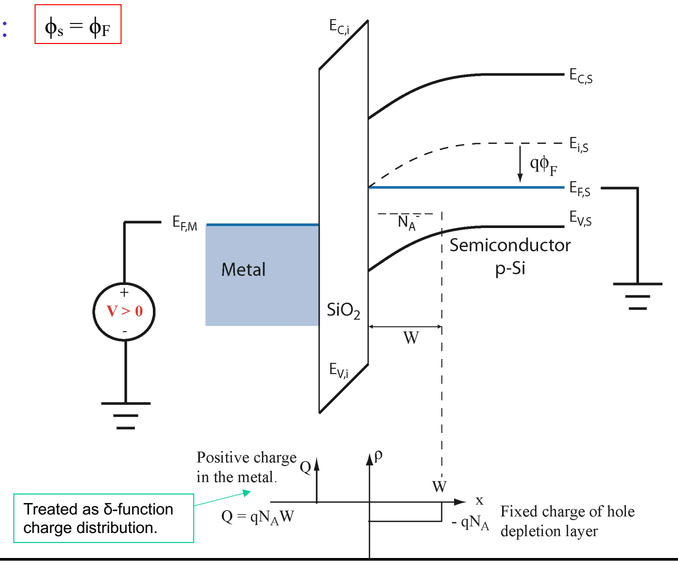
1. Flat Band is when the bands are flat in the semiconductor. i.e. the electric field in the semiconductor is zero. The field in the oxide does not have to be zero, although in this case, it was drawn that way.
2. The voltage applied to the metal gate to make this happen is the flat band voltage, V_{FB}.



16

MOS Depletion

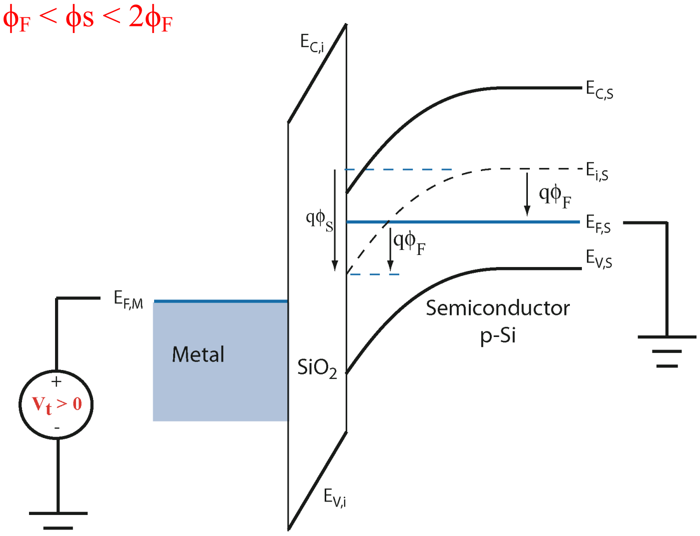
1. Depletion is when the majority carrier density is depleted at the surface of the Si. $\boxed{0 < \phi_s < \phi_F}$
2. There is a depletion layer of width W with charge ($-qN_A$).
3. Midgap: $\boxed{\phi_s = \phi_F}$



17

MOS Inversion

1. Inversion is when the carrier density at the surface of the Si is inverted so that $n_{surface} = p_{bulk}$.
2. This happens when $\phi_s = 2\phi_F$. $q\phi_s = E_i(\infty) - E_i(0) = 2q\phi_F$
3. Weak inversion: $\phi_F < \phi s < 2\phi_F$



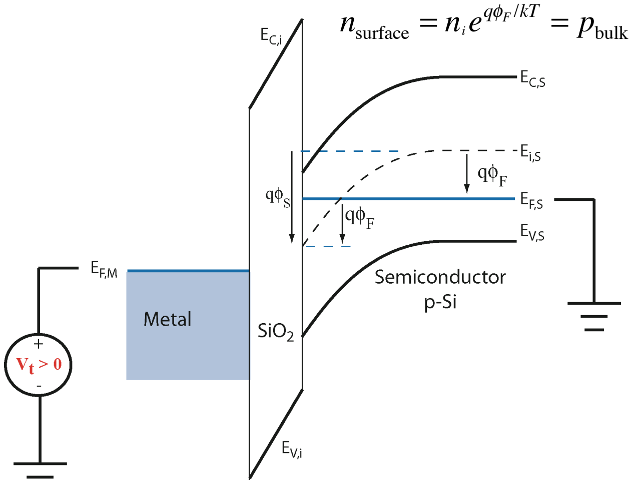
18

MOS Inversion

1. Inversion is when the carrier density at the surface of the Si is inverted so that $n_{surface} = p_{bulk}$.
2. This happens when $\phi_s = 2\phi_F$. $q\phi_s = E_i(\infty) - E_i(0) = 2q\phi_F$

$$n_{\text{surface}} = n_i e^{\left[\left(E_{\text{F,surface}} - E_{\text{i,surface}}\right)/kT\right]} = p_{\text{bulk}} = n_i e^{\left[\left(E_{\text{i,bulk}} - E_{\text{F,bulk}}\right)/kT\right]}$$

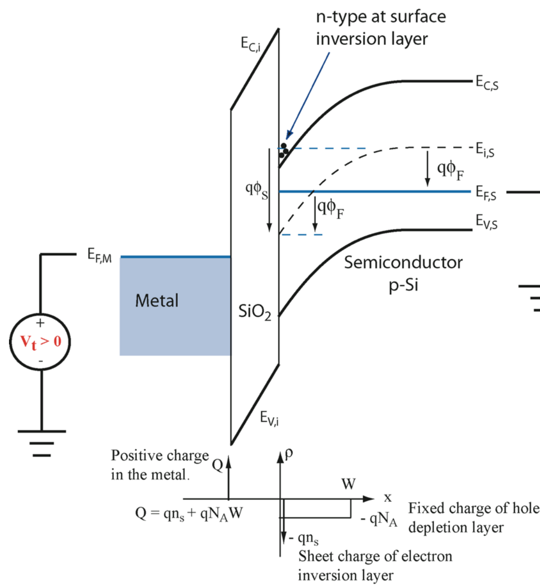
$$n_{\text{surface}} = n_i e^{q\phi_F/kT} = p_{\text{bulk}} = n_i e^{q\phi_F/kT}$$



19

MOS Inversion Charge Distribution

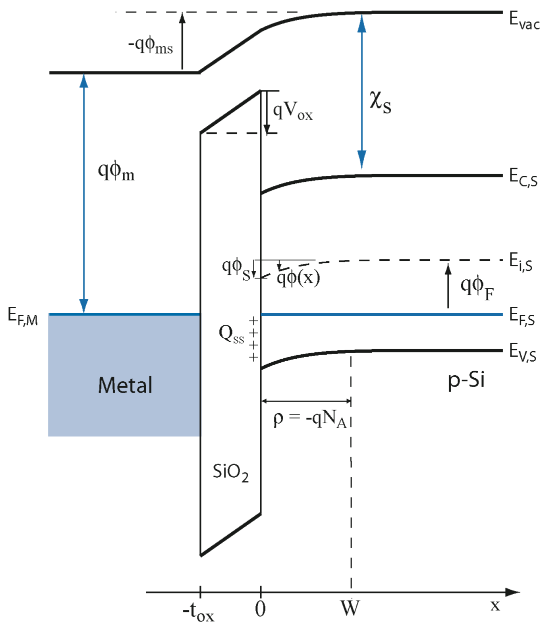
1. The inversion charge will be treated as a delta-function charge distribution at the surface (SiO2 / Si interface).
2. The negative depletion-layer charge is also still there.



20

MOS Electrostatics

1) $q\phi_m$ = work funcion of metal: $E_{vac} - E_{F,M}$.
2) χ_s = electron affinity of semiconductor: $\chi_s = E_{vac} - E_{C,S}$.
3) ϕ_{ms} = Metal-semiconductor work function difference.
   $\phi_{ms} = (E_{vac} - E_{F,M}) - (E_{vac,S} - E_{F,S})$
   $\phi_{ms} = \phi_m - (\chi + E_G/2q + \phi_F)$
4) $q\phi_F = E_{i,S} - E_{F,S} = (kT) \ln(N_A/n_i)$
5) Q_{SS} is fixed charge at the oxide – Si interface (x=0).
6) ϕ_s = surface potential of semiconductor. $\phi(\infty) = 0$. It is also the total voltage drop across the Si.
7) V_{ox} is the voltage drop across the oxide.

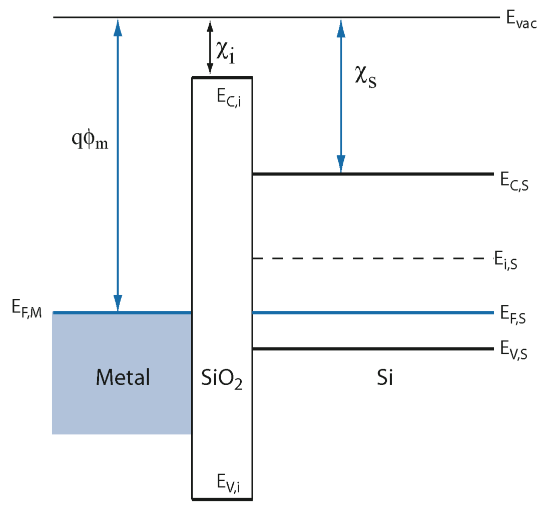


21

Ideal MOS Electrostatics

1) $\phi_{ms} = 0$.

2) $Q_{SS} = 0$.



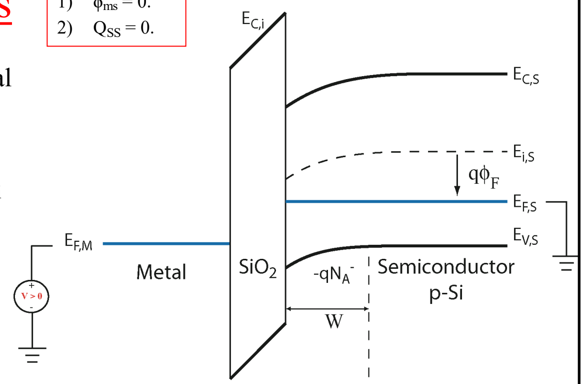
Equilibrium band diagram of ideal MOS capacitor. There is no voltage drop anywhere. Flatband.

22

Ideal MOS Electrostatics

1) $\phi_{ms} = 0$.
2) $Q_{SS} = 0$.

1) Apply a positive voltage to the metal to deplete holes at the Si surface.
2) Use depletion approximation.
3) At Si / SiO_2 interface, $\varepsilon_{ox}\mathcal{E}_{ox} = \varepsilon_{Si}\mathcal{E}_{Si}$
4) Determine W in terms of ϕ_s.

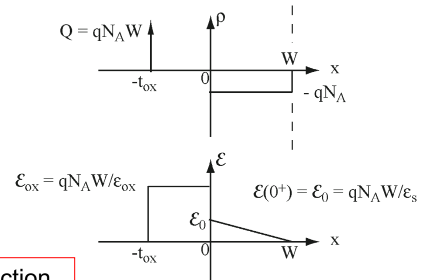


$$\frac{d\mathcal{E}}{dx} = \frac{\rho(x)}{\epsilon} \Rightarrow \int_{\mathcal{E}=0}^{\mathcal{E}(x)} d\mathcal{E} = \int_{-t_{ox}^-}^{x} dx' \frac{\rho(x')}{\epsilon}$$

$$\int_{V(0)}^{V(W)=0} dV = -V(0) = -\int_0^W dx\, \mathcal{E}(x) = -\frac{1}{2}\mathcal{E}_0 W$$

$$V(0) = \frac{1}{2}\mathcal{E}_0 W = \frac{qN_A W^2}{2\varepsilon_s}$$

$$\phi_s = \frac{qN_A W^2}{2\varepsilon_s}$$



$$\boxed{W = \sqrt{\frac{2\varepsilon_s \phi_s}{qN_A}}}$$

W of a one-sided n^+-p junction with ϕ_s replacing ($V_{bi} - V_A$) .

23

Ideal MOS Electrostatics

1) $\phi_{ms} = 0$.
2) $Q_{SS} = 0$.

$$\boxed{W = \sqrt{\frac{2\varepsilon_s \phi_s}{qN_A}}}$$

1) Total charge in the depletion region (Q_d):

$$\boxed{-qN_A W = -\sqrt{2qN_A\varepsilon_s\phi_s}\ \left(\frac{\mathrm{C}}{\mathrm{cm}^2}\right) \equiv Q_d}$$

2) Voltage drop across the oxide (V_{ox}):

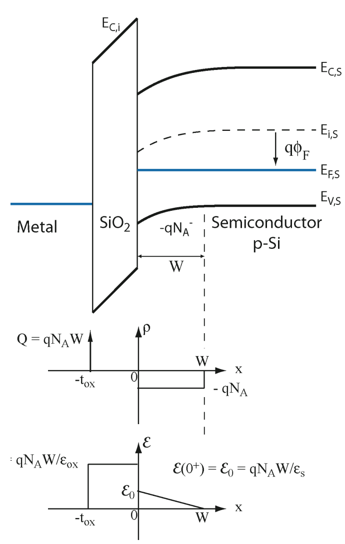
$$V_{ox} = \frac{qN_A W t_{ox}}{\varepsilon_{ox}} = \frac{qN_A W}{C_{ox}}$$

$$\boxed{C_{ox} \equiv \frac{\varepsilon_{ox}}{t_{ox}}}$$

Oxide capacitance (F/cm^2)

$$\boxed{V_{ox} = \frac{-Q_d}{C_{ox}}}$$

($Q_d < 0$)



3) Gate Voltage (V_G): $V_G = V_{ox} + V_{Si} = V_{ox} + \phi_s$ (ϕ_s is the total voltage drop across the Si.)

24

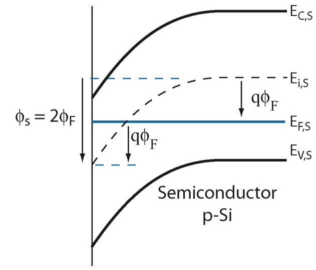
Ideal MOS Electrostatics – Inversion (Threshold)

1) $\phi_{ms} = 0$.
2) $Q_{SS} = 0$.

- Inversion occurs when $\phi_s = 2\phi_F$.
- The applied gate voltage at which this occurs is the threshold voltage, V_t.
- At inversion, the surface charge density is inverted, i.e. $n(x=0) = p(x=\infty)$ (for p-type semiconductor).

$$n_{\text{surface}} = n_i e^{q\phi_F/kT} = p_{\text{bulk}} = n_i e^{q\phi_F/kT} = N_A$$

$$\phi_F = \frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$



- At Inversion (Threshold)

$$W_t = \sqrt{\frac{2\varepsilon_s\phi_s}{qN_A}} = \sqrt{\frac{2\varepsilon_s 2\phi_F}{qN_A}} = 2\sqrt{\frac{\varepsilon_s\phi_F}{qN_A}}$$

where

$$\phi_F = \frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$

$$Q_{d,t} = -2\sqrt{qN_A\varepsilon_s\varphi_F}\left(\frac{\text{C}}{\text{cm}^2}\right)$$

$$\boxed{V_{G,t} = \frac{-Q_{d,t}}{C_{ox}} + 2\phi_F}$$

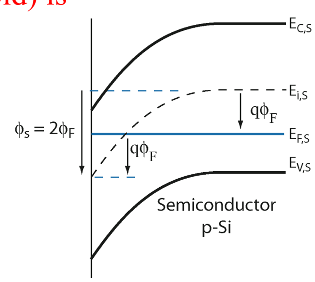
25

Ideal MOS Electrostatics - Inversion

1) $\phi_{ms} = 0$.
2) $Q_{SS} = 0$.

- The threshold voltage (the gate voltage at threshold) is

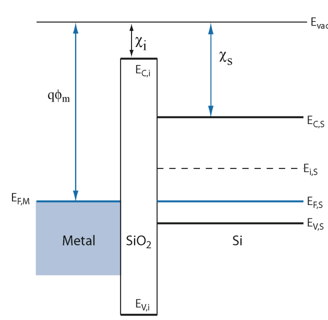
$$\boxed{V_{G,t} = \underbrace{\frac{-Q_{d,t}}{C_{ox}}}_{\text{Drop across the oxide}} + \underbrace{2\phi_F}_{\text{Drop across the Si}}}$$



where

$$Q_{d,t} = -2\sqrt{qN_A\varepsilon_s\phi_F}\left(\frac{\text{C}}{\text{cm}^2}\right)$$

- This is the voltage starting from the ideal flat-band condition shown at right.



26

MOS Electrostatics - Inversion

- For non-ideal MOS with $\phi_{ms} \neq 0$ and $Q_{ss} \neq 0$, we have to apply a gate voltage V_{FB} to get to flat-band. Once at flat-band, everything proceeds as before so that the threshold voltage is

$$\boxed{V_{G,t} = V_{FB} + 2\phi_F - \frac{Q_{d,t}}{C_{ox}}}$$

V_G needed to get to flatband.

Drop across the Si

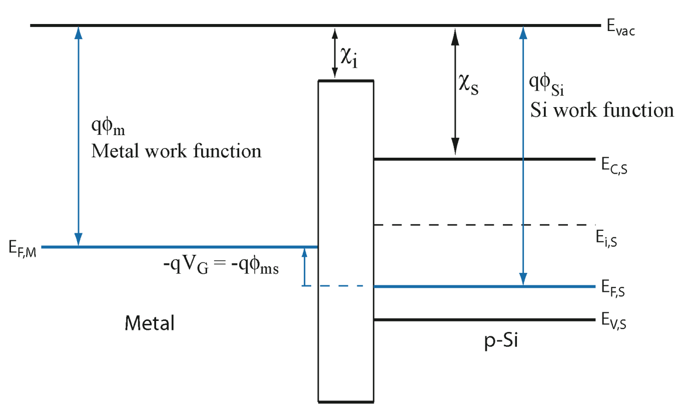
Drop across the oxide

- What is the flat-band voltage?

27

MOS Electrostatics – Flat-band-Voltage – non-zero ϕ_{ms}

- Consider $\phi_{ms} \neq 0$, i.e. the metal-semiconductor work-function difference is non-zero. $q\phi_{ms} = q\phi_m - q\phi_{Si} < 0$



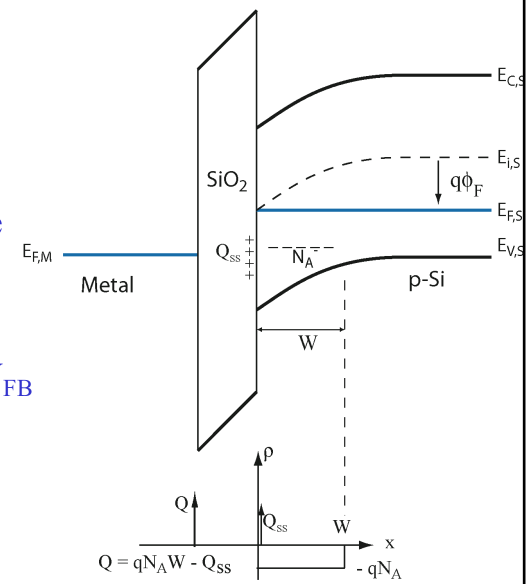
- You have to apply a gate voltage $V_G = \phi_{ms}$ to get to flat band.

28

MOS Electrostatics – Flat-band-Voltage – non-zero Q_{ss}

- Consider $Q_{ss} \neq 0$, i.e. fixed charge at x=0 (SiO_2 / Si interface).
- Qss gives rise to an extra voltage drop across the oxide of $(-Q_{ss} / C_{ox})$.
- Thus, the flat-band voltage V_{FB} is:

$$\boxed{V_{FB} = \phi_{ms} - \frac{Q_{ss}}{C_{ox}}}$$



29

MOS Electrostatics - Threshold

- For non-ideal MOS with $\phi_{ms} \neq 0$ and $Q_{ss} \neq 0$, the threshold voltage is

$$\boxed{V_{G,t} = V_{FB} + 2\phi_F - \frac{Q_{d,t}}{C_{ox}}}$$

$$\boxed{V_{FB} = \phi_{ms} - \frac{Q_{ss}}{C_{ox}}}$$

$$\boxed{V_{G,t} = \phi_{ms} + 2\phi_F - \frac{Q_{ss}}{C_{ox}} - \frac{Q_{d,t}}{C_{ox}}}$$

30

MOS Electrostatics - Threshold

- For non-ideal MOS with $\phi_{ms} \neq 0$ and $Q_{ss} \neq 0$, the threshold voltage is

$$\boxed{V_{G,t} = \phi_{ms} + 2\phi_F - \frac{Q_{ss} + Q_{d,t}}{C_{ox}}}$$

V_G needed overcome the workfunction difference.

Drop across the Si

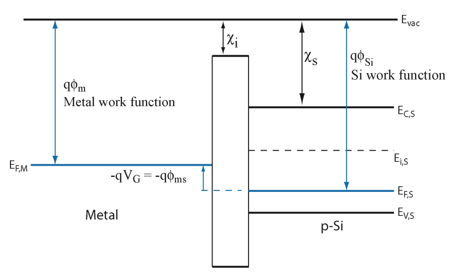
Drop across the oxide

31

MOS Electrostatics - Threshold

- Summary

$$V_{G,t} = \phi_{ms} + 2\phi_F - \frac{Q_{ss}}{C_{ox}} - \frac{Q_{d,t}}{C_{ox}}$$



$$V_{G,t} = V_{FB} + 2\phi_F - \frac{Q_{d,t}}{C_{ox}}$$

$$V_{FB} = \phi_{ms} - \frac{Q_{ss}}{C_{ox}}$$

$$\phi_{ms} = \phi_m - \frac{1}{q}\left[\chi + (E_C - E_{F,S})_{x=\infty}\right]$$

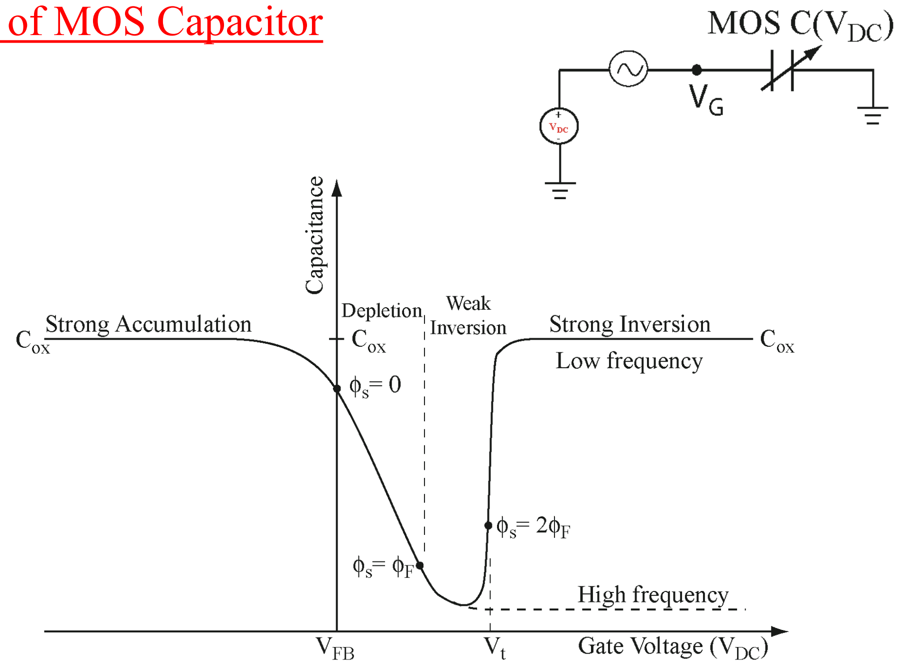
$$\phi_F = \frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$

$$Q_{d,t} = -2\sqrt{qN_A\varepsilon_s\phi_F} \quad \left(\frac{\mathrm{C}}{\mathrm{cm}^2}\right)$$

$$W_t = 2\sqrt{\frac{\varepsilon_s\phi_F}{qN_A}}$$

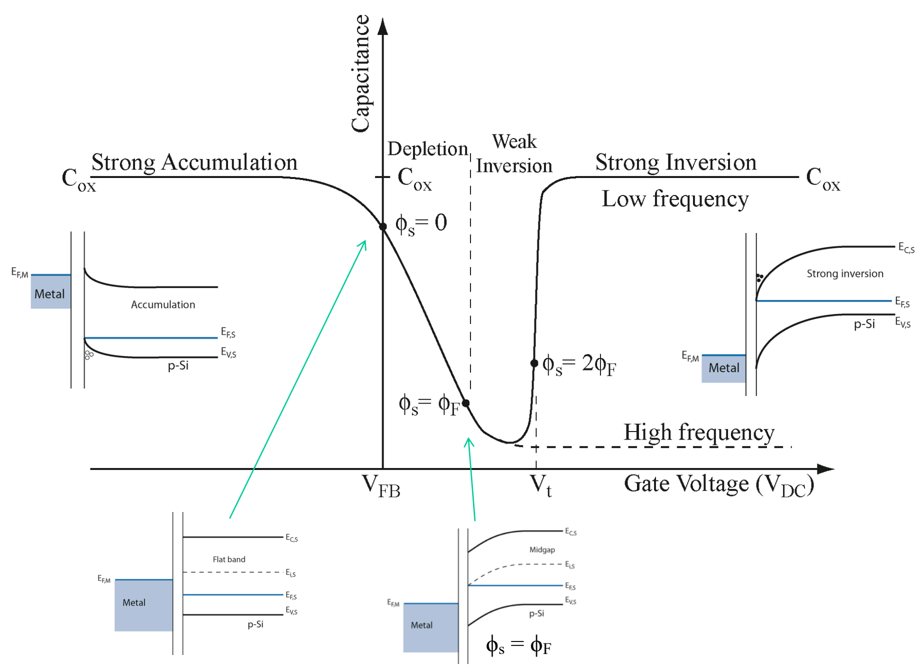
32

C-V of MOS Capacitor



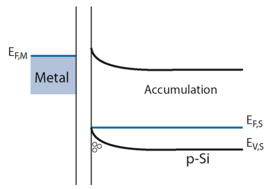
33

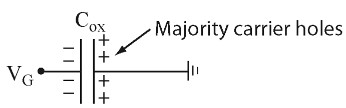
C-V of MOS Capacitor

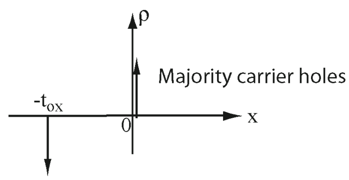


34

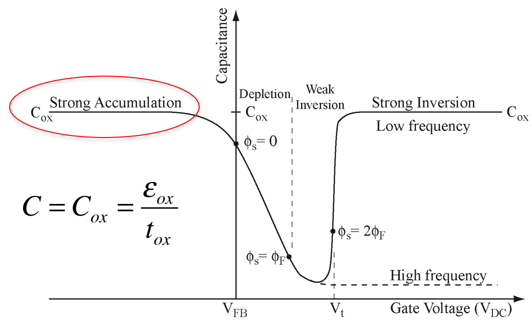
MOS Capacitor in Accumulation







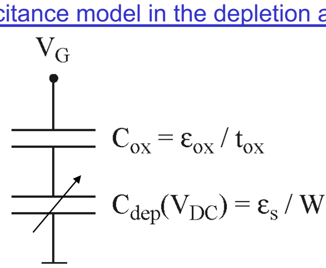
$$C = C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

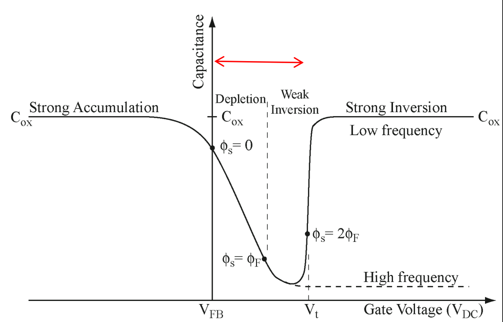


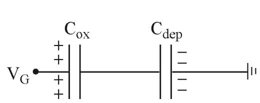
35

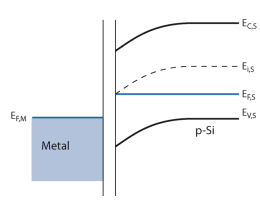
MOS Capacitor in Depletion / Weak-Inversion

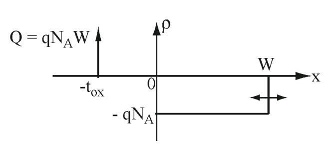
Capacitance model in the depletion and weak inversion region:

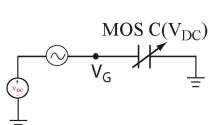








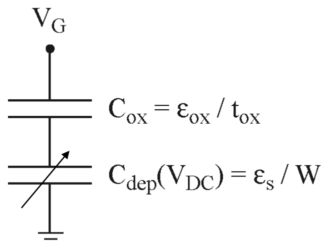


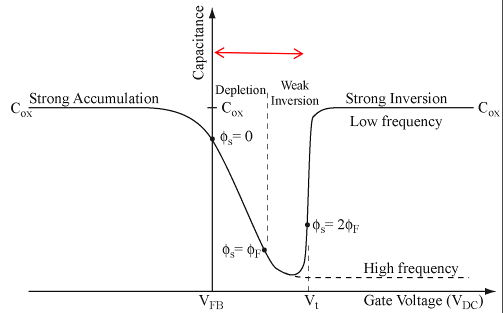


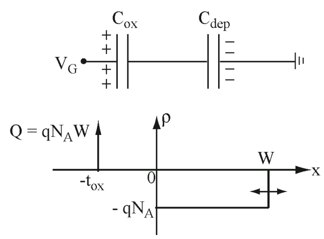
36

MOS Capacitor in Depletion / Weak-Inversion

Capacitance model in the depletion and weak-inversion region:





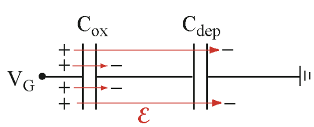


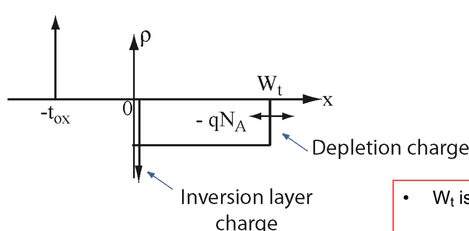
$$C_{depl} = \frac{C_{ox}C_d}{C_{ox}+C_d} = \frac{C_{ox}}{1+C_{ox}/C_d} = \frac{C_{ox}}{1+\dfrac{\varepsilon_{ox}}{\varepsilon_s}\dfrac{W_d}{t_{ox}}}$$

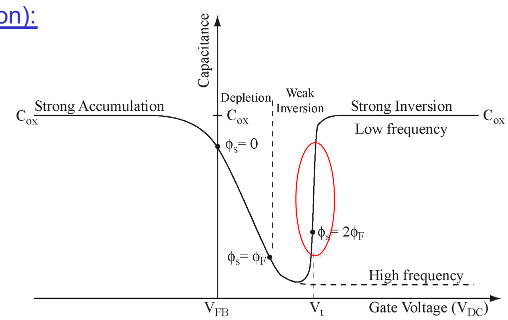
37

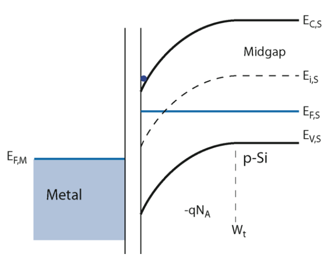
Low-freqency MOS Capacitor at Inversion (Threshold)

Capacitance at threshold (the point of inversion):

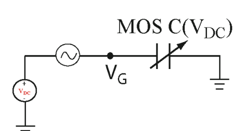






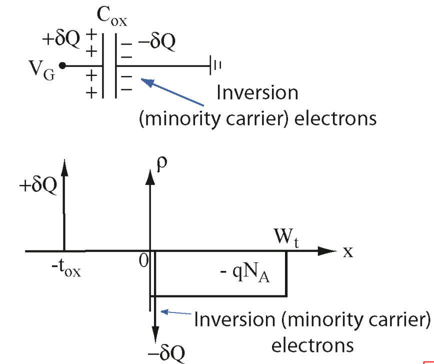


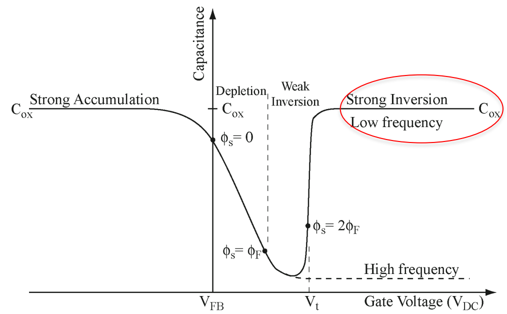
- W_t is the width of the depletion region at threshold.
- Under steady-state (DC) or very low frequency, W_t is the maximum width of W.
- For $V_G > V_t$, the additional ΔQ on the metal gate is imaged by the additional ΔQ in the inversion layer.

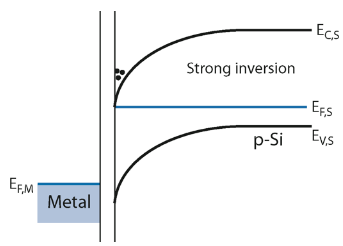


38

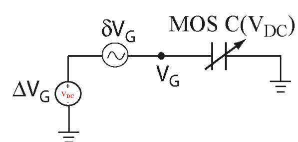
Low-freqency MOS Capacitor in Strong Inversion





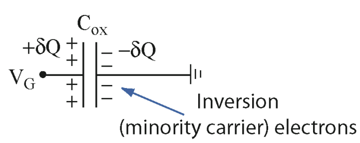


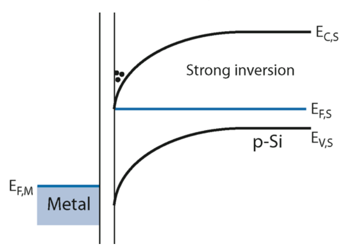
- In low-frequency, strong inversion, all of the +δQ on the metal is imaged on the –δQ of the inversion layer at the Si-SiO2 interface.
- The inversion layer is minority-carriers.

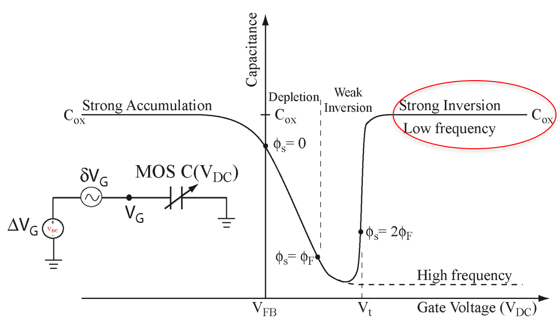


39

Low-freqency MOS Capacitor in Strong Inversion



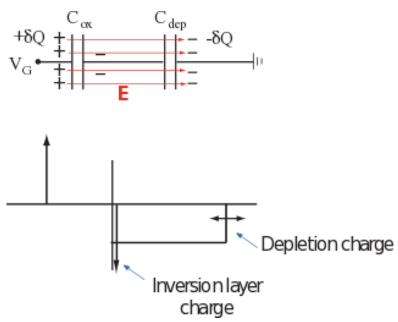


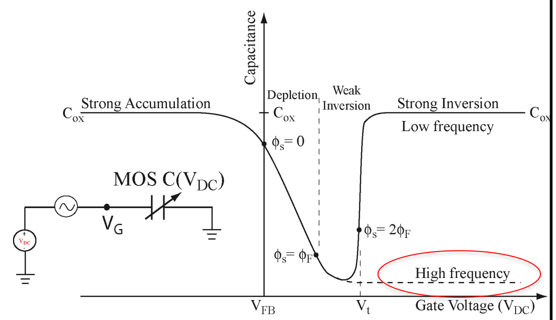


- In low-frequency, strong inversion, all of the +δQ on the metal is imaged on the –δQ of the inversion layer at the Si-SiO_2 interface.
- The inversion layer is a layer of minority-carriers.
- Where do these minority-carriers come from?

40

High-freqency MOS Capacitor in Strong Inversion

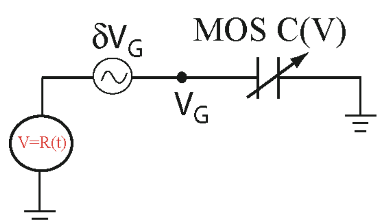


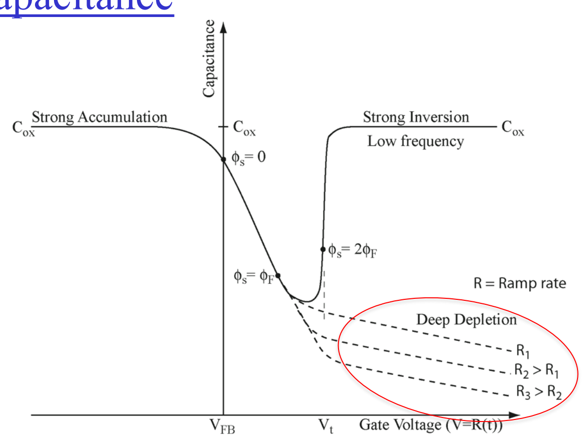


- The source of the minority-carrier inversion layer is thermal generation.
- This is governed by the minority-carrier lifetime τ_n, which we know is relatively long.
- The rate at which minority-carriers can respond is ~ $1/\tau_n$ which is slow.
- If the frequency (f) of the ac source >> $1/\tau_n$, the inversion layer cannot respond.
- Then the $+\delta Q$ on the metal is imaged by the $-\delta Q$ at the edge of the depletion region, W_t. This $-\delta Q$ is the result of the majority-carrier holes moving at the edge of the depletion region.
- Thus, C remains at its minimum determined by the maximum depletion $W = W_t$.

41

Deep-depletion MOS Capacitance



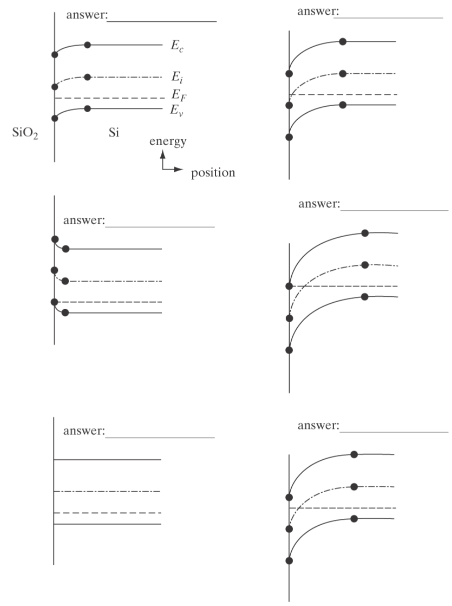


- In deep depletion, you sweep the biasing voltage too fast for the depletion layer to form.
- Push the depletion width out to $W > W_t$.
- Capacitance keeps decreasing.
- If you then hold V at $V > V_t$, the depletion layer will eventually form, and C will rise to its high-frequency value.

42

Retreival

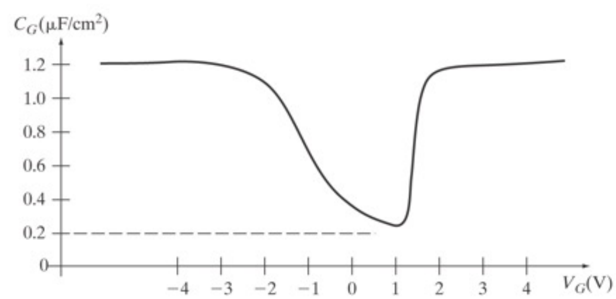
- Determine whether the following MOS capacitor band diagrams correspond to accumulation, weak inversion, depletion, strong inversion, flat band, or threshold.



43

Retreival

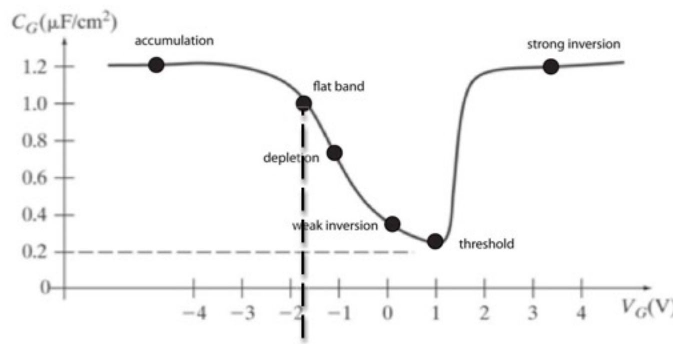
- On the Si / SiO2 CV curve, label the regions corresponding to (a) weak inversion, (b) flat band, (c) strong inversion, (d) accumulation, (e) threshold, and (f) depletion.



44

Retreival

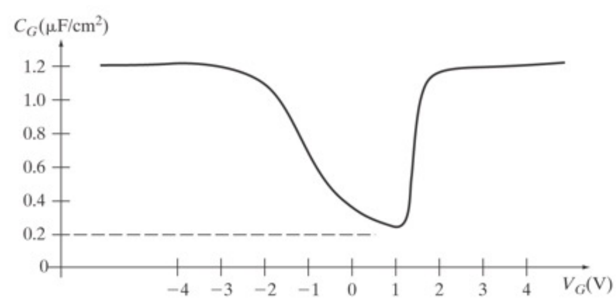
- On the Si / SiO2 CV curve, label the regions corresponding to (a) weak inversion, (b) flat band, (c) strong inversion, (d) accumulation, (e) threshold, and (f) depletion.



45

Retreival

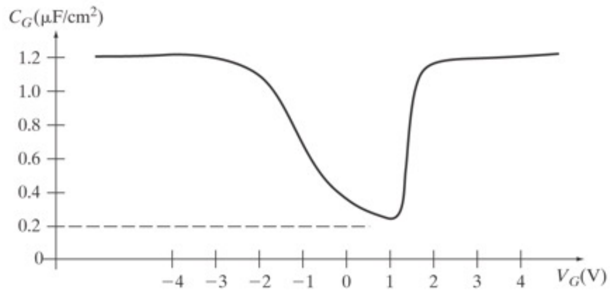
- Is this a low frequency or high frequency CV curve. Explain.



46

Retreival

- Is this a low frequency or high frequency CV curve. Explain.

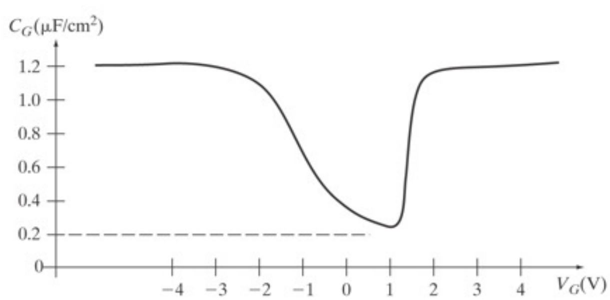


Low frequency. C returns to C_{ox} in inversion. The inversion layer has time to respond to the ac signal.

47

Retreival

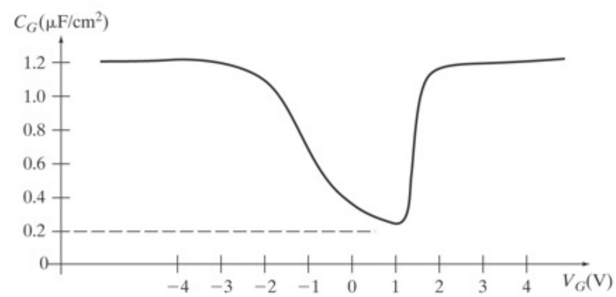
- If the metal-semiconductor work function difference is zero, $\varphi_{ms} = 0$, what is Qss? Show work.



48

Retreival

- If the metal-semiconductor work function difference is zero, φms = 0, what is Qss? Show work.



$V_{FB} = \varphi_{ms} - Q_{ss}/C_{ox}$.

$V_{FB} \approx -1.8V = -Q_{ss}/C_{ox}$

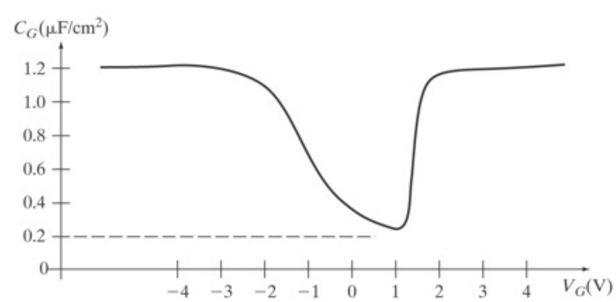
$Q_{ss} = 1.8\,C_{ox} = 1.8 \times 1.2\times10^{-6} = 2.16\times10^{-6}$ C/cm^2

The density of surface states is Q_{SS} / 1.602e-19 = 1.35×10^{13} cm^{-2}.

49

Retreival

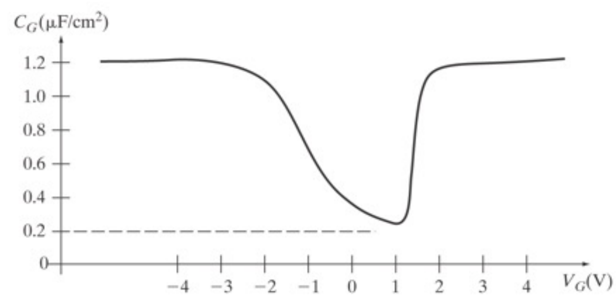
- If $Q_{ss} = 0$, what is φ_{ms}? Show work.



50

Retreival

- If $Q_{ss} = 0$, what is φ_{ms}? Show work.



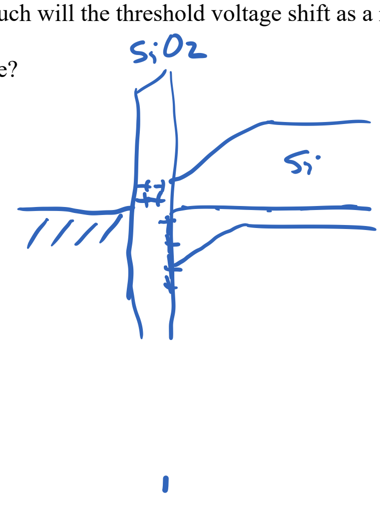
$V_{FB} = \phi_{ms} - Q_{ss}/C_{ox}$.

If Qss = 0, what is ϕ_{ms}? Show work. From above equation for V_{FB}, ϕ_{ms} = -1.8V.

51

Retreival

- Start with an ideal Si / SiO_2 MOS capacitor. How much will the threshold voltage shift as a result of
  (a) A uniform distribution of charge, ρ_{ox}, in the oxide?
  (b) A sheet of charge Q_s at the metal-SiO_2 interface.



52

Retreival

- Start with an ideal Si / SiO_2 MOS capacitor. How much will the threshold voltage shift as a result of
  (a) A uniform distribution of charge, ρ_{ox}, in the oxide?
  (b) A sheet of charge Q_s at the metal-SiO_2 interface.

A uniform charge distribution in the oxide will give rise to a voltage drop in the oxide of $(1/2)\ \rho_{ox}\ t_{ox}^2 / \varepsilon_{ox}$ and a corresponding threshold voltage shift of $-(1/2)\ \rho_{ox}\ t_{ox}^2 / \varepsilon_{ox}$ (for NMOS). The total charge per unit area in the oxide is $Q_{ox} = \rho_{ox}\ t_{ox}$. Thus, the shift is $-(1/2)\ Q_{ox} / C_{ox}$ where $C_{ox} = \varepsilon_{ox} / t_{ox}$. This is ½ of the shift that would result if all of Q_{ox} were at the SiO2 / Si interface.

53

Retreival

- Start with an ideal Si / SiO_2 MOS capacitor. How much will the threshold voltage shift as a result of
  (a) A uniform distribution of charge, ρ_{ox}, in the oxide?
  (b) A sheet of charge Q_s at the metal-SiO_2 interface.

(b) A sheet of charge Q_s at the metal-SiO_2 interface. Causes no voltage drop across the oxide and no threshold voltage shift. The easiest way to see this is in terms of capacitance. The capacitance is $C = \varepsilon_{ox} / t$ with t going to zero. Thus, $\Delta V = \Delta Q/C = 0$.

54

- Go to MOSFETs

55